

The empirical recurrence for OEIS A256026, recovered from its tail

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Abstract

OEIS A256026 records “Empirical recurrence of order 77” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 105$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 80$. Its last coefficient is $c_{77} = 20 \neq 0$, so no further tail satisfies anything shorter; any order-77 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A256026 is “Number of $(n+2) \times (5+2)$ 0..1 arrays with no 3×3 subblock diagonal sum 0 or 1 and no antidiagonal sum 0 or 1 and no row sum 1 and no column sum 1.”. It has offset 1 and begins

798, 3500, 18881, 96363, 494704, 2608924, 13744343, 72433360, ...

The entry states:

Empirical recurrence of order 77 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 11:44:01, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 105$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 105$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 77: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 77.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 210$ exact terms from $n = 3$ on, it returns order 77 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{77} c_i a(n-i),$$

c_1	7	c_{21}	−18289248	c_{41}	−1546468583	c_{61}	−3732992
c_2	−17	c_{22}	16799667	c_{42}	−1629683050	c_{62}	−1943961
c_3	70	c_{23}	74701382	c_{43}	−725507124	c_{63}	349255
c_4	−91	c_{24}	100840730	c_{44}	356625384	c_{64}	1137737
c_5	−209	c_{25}	32537531	c_{45}	910214368	c_{65}	720900
c_6	−390	c_{26}	−139247277	c_{46}	734476552	c_{66}	100590
c_7	−2376	c_{27}	−287453152	c_{47}	196090393	c_{67}	−164146
c_8	6527	c_{28}	−240337167	c_{48}	−224964609	c_{68}	−138794
c_9	14970	c_{29}	106252789	c_{49}	−299861820	c_{69}	−40073
c_{10}	20724	c_{30}	547568137	c_{50}	−132440197	c_{70}	8631
c_{11}	5177	c_{31}	679053969	c_{51}	47165128	c_{71}	12365
c_{12}	−201462	c_{32}	228750707	c_{52}	103051747	c_{72}	3991
c_{13}	−336672	c_{33}	−668631501	c_{53}	52216367	c_{73}	−156
c_{14}	−242975	c_{34}	−1283337997	c_{54}	−13551143	c_{74}	−344
c_{15}	678708	c_{35}	−1045975325	c_{55}	−37202230	c_{75}	26
c_{16}	2916911	c_{36}	126378307	c_{56}	−20631704	c_{76}	96
c_{17}	3535080	c_{37}	1434530358	c_{57}	2208584	c_{77}	20
c_{18}	175194	c_{38}	1885751330	c_{58}	10676806		
c_{19}	−10753190	c_{39}	1108241584	c_{59}	5629206		
c_{20}	−22917379	c_{40}	−413961983	c_{60}	−1428667		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 80$, and it is the recurrence OEIS A256026 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{77} - c_1 x^{76} - \dots - c_{77}$. Its constant term is $-c_{77} = -20$, which is not zero, so $q_{\min}(0) \neq 0$ and

$q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 77 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 77, so $\deg q = 77$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 77, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 77, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 20.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-77 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A256026.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.